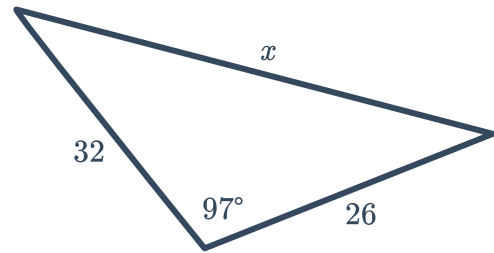


Perimeter from radial surveys



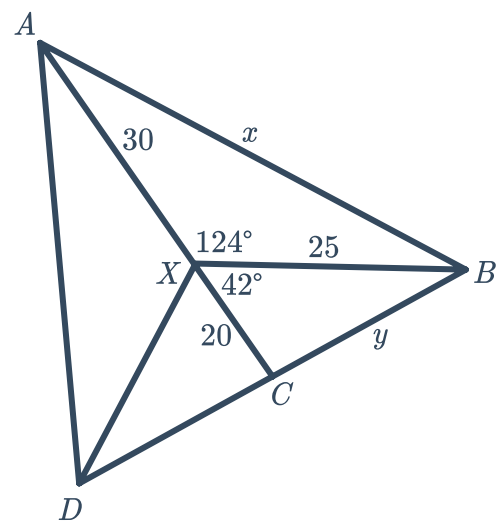
- 1 Consider the following triangle, taken from a field diagram. Note that all measurements are in metres.

- Find the side length, x , correct to two decimal places.
- Hence, find the perimeter of the triangle correct to two decimal places.



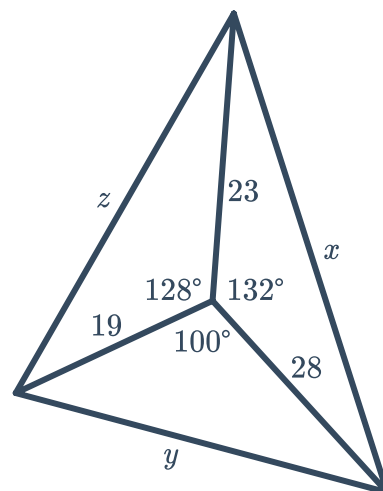
- 2 Consider the following diagram of a field. Note that all measurements are in metres.

- Find the value of x , correct to two decimal places.
- Find the value of y , correct to two decimal places.
- The lengths of CD and AD are 29.02 m and 44.89 m respectively. Find the perimeter of triangle ABD .



- 3 Consider the following field diagram. Note that all measurements are in metres.

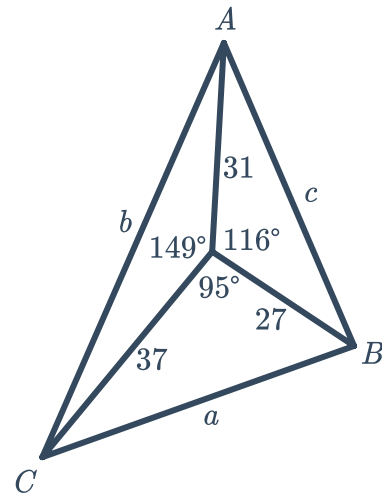
- Find the value of x , correct to two decimal places.
- Find the value of y , correct to two decimal places.
- Find the value of z , correct to two decimal places.
- Find the perimeter of the whole field correct to two decimal places.



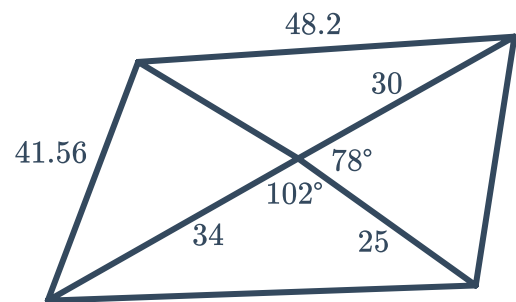
- 4 Consider the following diagram. Note that all measurements are in metres.



- Find the value of c , correct to two decimal places.
- Find the value of a , correct to two decimal places.
- Find the value of b , correct to two decimal places.
- Hence, find the perimeter of the field.

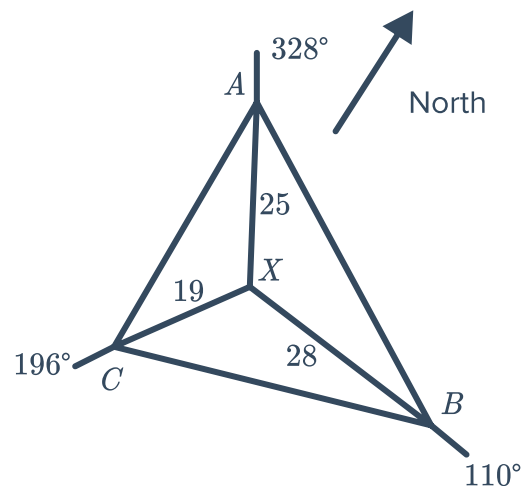


- 5 Consider the following field diagram. Find the perimeter of the whole field correct to the nearest metre.

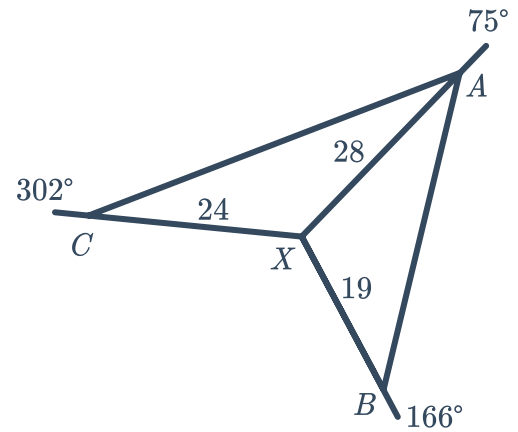


- 6 Consider the following diagram. Points A , B and C lie at true bearings of 328° , 110° and 196° respectively from point X . Note that all measurements are in metres.

- Find the size of $\angle AXB$.
- Find the size of $\angle BXC$.
- Find the size of $\angle CXA$.
- Find the perimeter of the field correct to two decimal places.

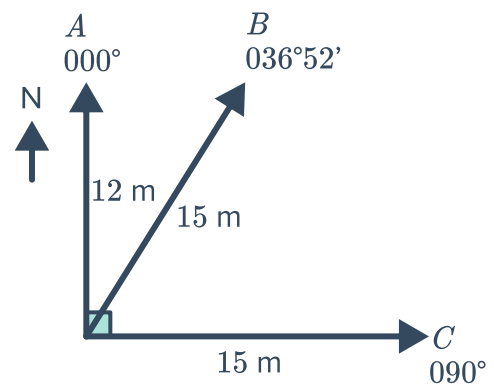


- 7 Consider the following field diagram. Points A , B and C lie at true bearings of 75° , 166° and 302° respectively from point X . Note that all measurements are in metres. Find the perimeter of the field diagram, correct to two decimal places.



- 8 Consider the following radial survey of a block of land. In the survey, B is directly east of A .

- Draw a diagram of the block of land from the radial survey.
- Calculate the length of AB to the nearest metre.
- Calculate the length of BC to the nearest metre.
- Hence calculate the perimeter of the block of land.

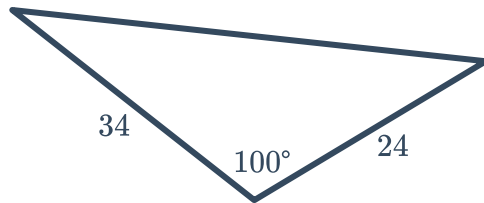




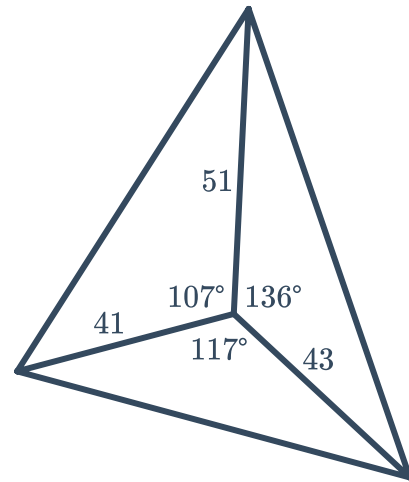
Area from radial surveys

- 9 Find the area of the following fields correct to two decimal places. Note that all measurements are in metres.

a

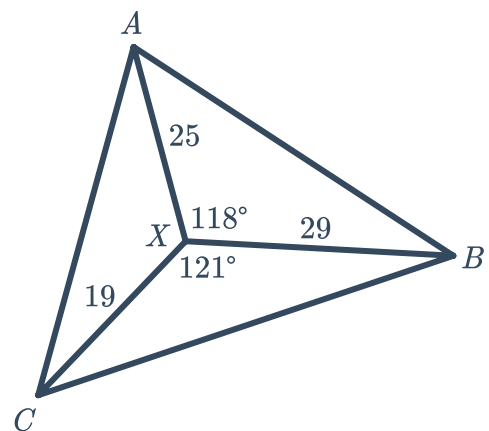


b

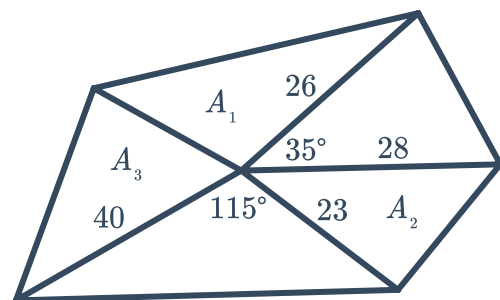


- 10 Consider the following field diagram. Note that all measurements are in metres.

- a Find the size of $\angle AXC$.
b Find the area of the entire field, correct to two decimal places.



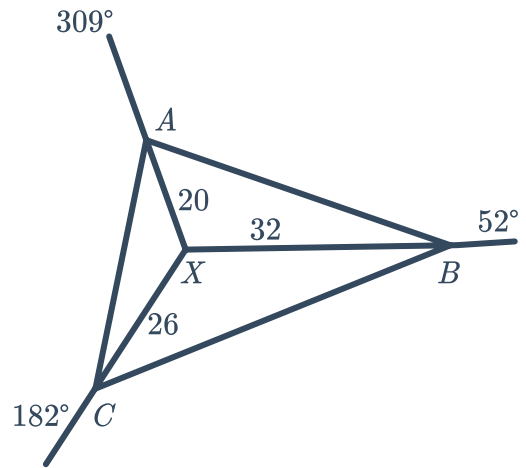
- 11 In the field diagram below, $A_1 = 275.23 \text{ m}^2$, $A_2 = 170.63 \text{ m}^2$ and $A_3 = 416.90 \text{ m}^2$. Find the area of the whole field correct to two decimal places.



and C lie at true bearings of 309° , 52° and 182° respectively from point X , where $AX = 20$, $BX = 32$ and $CX = 26$. Note that all measurements are in metres.



- a Find the size of angles $\angle AXB$, $\angle AXC$ and $\angle CXB$.
- b Find the total area of the field, correct to two decimal places.



- 13 Consider the following four sided field diagram. The four lines passing from the centre to the corners of the field are at true bearings of 46° , 111° , 233° and 311° . Note that all measurements are in metres. Find the area of the field correct to two decimal places.

